

THE ABERDARE TECHNICAL COLLEGE
LEARNING PLAN – COMPUTER OPERATIONS
1.INSTITUTION DETAILS

Item	Details
MODULE	MODULE 1
UNIT OF COMPETENCE	COMPUTER OPERATIONS
UNIT CODE	0611 351 02A
TRAINER	JAMES NDIRANGU
COMPETENCE LEVEL	LEVEL 5
INSTITUTION	THE ABERDARE TECHNICAL COLLEGE
NUMBER OF LEARNERS	1
DATE OF PREPARATION	04/09/2026
NUMBER OF LEARNERS	MALE.....1.....FEMALE.....0.....
TRAINING APPROACH	DEMONSTRATION, GUIDED PRACTICE, DISCUSSION, INDIVIDUAL PRACTICAL EXERCISES AND ASSESSMENT

SKILL OR JOB TASK

Perform computer operations by processing word documents, manipulating spreadsheets, maintaining computerized databases, preparing presentations, manipulating graphic applications, performing document production, and participating in online and mobile collaboration.

1. BENCHMARK OR CRITERIA TO BE USED

1. Process computerized word document

- 1.1 Ergonomics risk factors observed as per work place procedures*
- 1.2 Word document is created as per work requirements*
- 1.3 Tables are created and manipulated as per work requirements*
- 1.4 Mail merging is performed as per work requirements*
- 1.5 Word processing Objects are inserted as per user requirements*
- 1.6 List of figures and table of content are generated as per user requirements*
- 1.7 Computerized Word documents are printed as per workplace procedure.*

2. Manipulate computerized spreadsheets

- 2.1 Spreadsheet workbook is created as per work requirements*
- 2.2 Cell referencing is performed as per task requirements*
- 2.3 Formula and functions are applied as per work requirements*
- 2.4 Charts are generated as per work requirements*

3. Maintain computerized database

3.1 Computerized database user requirements are collected as per work requirements.

3.2 Computerized database schemas are designed as per task requirements.

3.3 Creation of Computerized database objects as per task requirements.

3.4 Data manipulation is performed as per task requirements.

4. Prepare Power point presentation

4.1 PowerPoint Presentation requirements are collected as per work requirements

4.2 PowerPoint slides are created as per work requirement

4.3 Presentation views are exhibited as per work requirements.

4.4 Animations and transitions are performed as per work requirements

4.5 PowerPoint slides are manipulated as per work requirements

4.6 PowerPoint slides are printed as per work requirements

5. Manipulate graphic application

5.1 Identifying graphic design requirements

5.2 Graphic design created as per task requirements

5.3 Graphic design published as per the task requirements

6. Perform document production

6.1 Document is printed as per user specifications

6.2 Documents are scanned as per user specifications

6.3 Documents are duplicated as per user specifications

7. Perform Online Collaboration

7.1 Identification of Online collaboration tools as per the task requirements

7.2 Prepare online collaboration as per the task requirements.

7.3 Apply online collaborative tools as per the task requirements.
 Demonstrating Mobile collaborations as per task requirements

2.LEARNING PLAN

W k	Main Learning Outcome	Session Title / Detailed Subtopics	Learning Outcome	Trainer Activities	Trainee Activities	Resources and References	Assessment Methods	Reflections & Dates
1	Process computeriz ed word document	Session 1: Ergonomics & Introduction to Word Processing Subtopics: 1.1 Ergonomic risk factors & posture 1.2.1 Intro to word processing 1.2.2 Types of word processors (MS Word, LibreOffice, WPS, Google Docs)	By the end of the session, trainee should be able to: 1. Identify workstation ergonomic risks. 2. Explain posture importance. 3. List major word processing applications. 4. Differentiate local vs cloud word processors.	1. Demonstrates ergonomic workstation setup (chair height, monitor level, wrist support). 2. Explains RSI and eye strain risks. 3. Introduces word processing software types. 4. Leads comparative discussion.	1. Adjusts workstation setup following ergonomic principles. 2. Identifies potential workplace ergonomic hazards. 3. Launches and explores word processors. 4. Tabulates differences between office suites.	• Desktop Computers & Adjustable Workstations • Overhead Projector & Ergonomic Charts • MS Word 2019 / LibreOffice Writer • Ref: ISO 9241 Ergonomics Standard • Ref: TVET CDACC ICT Level 5 Manual	• Direct Observation Checklist • Oral Questioning on Ergonomics • Diagnostic Identification Quiz	Date: 07/09/2026 Reflections: Trainee demonstrate d good awareness of ergonomic posture. Hands-on setup successfully completed.
1	Process computeriz ed word document	Session 2: Creating, Editing & Formatting Basics Subtopics: 1.2.3 Creating word documents 1.2.4 Editing & formatting fundamentals 1.2.5 Text & paragraph editing (cut, copy, paste, find/replace)	By the end of the session, trainee should be able to: 1. Create and save new documents. 2. Perform text insertion and editing. 3. Apply paragraph alignment & spacing. 4. Utilize Find	1. Demonstrates file creation, naming conventions, and file formats (.docx, .pdf). 2. Guides keyboard shortcuts for editing (Ctrl+C, Ctrl+V, Ctrl+Z). 3. Assigns text formatting & layout editing exercise.	1. Creates, saves, and structures a multi- paragraph file. 2. Performs text editing operations. 3. Replaces target text using Find & Replace. 4. Applies line spacing (1.5,	• PC / Laptop with MS Office installed • Sample Unformatted Raw Text Files • Video Tutorial: Keyboard Shortcuts • Ref: Microsoft Office Specialist (MOS) Word Study Guide • Ref: CDACC Level 5 Trainer	• Practical Document Editing Exercise • Task Completion Timing Check • Formative Product Evaluation	Date: 09/09/2026 Reflections: Trainee mastered short-cut keys quickly. Good progress on file organization.

			and Replace tools.		double) and alignment.	Guide		
2	Process computerized word document	Session 3: Advanced Word Document Formatting Subtopics: 1.2.6 Word document formatting features 1.2.6.1 Text formatting (fonts, styles, effects) 1.2.6.2 Paragraph formatting (indents, bullet lists) 1.2.6.3 Page setup & borders	By the end of the session, trainee should be able to: 1. Apply advanced font formatting. 2. Customize indents, tabs, and line spacing. 3. Insert page borders, watermarks, and shading. 4. Format an entire document layout.	1. Demonstrates advanced typography settings (kerning, text effects). 2. Illustrates tab stops (Left, Right, Decimal) and bullet customization. 3. Issues a complex document formatting specification sheet.	1. Formats provided raw text to match target template. 2. Configures custom tab stops and bullet icons. 3. Adds page shading, borders, and section breaks. 4. Reviews visual appeal against standards.	• Workstation Computers & Projector • Style Guide Document Specifications • Sample Corporate Template Files • Ref: Shelly Cashman Series - Microsoft Office 365 & Word 2019 • Ref: Aberdare Tech ICT Dept Formatting Rules	• Practical Formatting Assessment • Rubric-based Document Quality Review	Date: 14/09/2026 Reflections: Formatting criteria met. Tab stops needed brief extra guidance.
2	Process computerized word document	Session 4: Enhancing Word-Processing Productivity Subtopics: 1.2.7.1 System options & preferences 1.2.7.2 Built-in Help resources & documentation 1.2.7.3 Magnification/zoom & view modes 1.2.7.4 Customizing Quick Access Toolbar & Ribbon	By the end of the session, trainee should be able to: 1. Configure Word application options. 2. Use built-in Help (F1) to solve software tasks. 3. Customize toolbars and view settings. 4. Manage document properties.	1. Demonstrates Word Options dialog box (AutoCorrect, Save interval). 2. Guides search techniques within MS Office Help. 3. Shows ribbon customization and QAT additions. 4. Assigns troubleshooting scenario.	1. Configures AutoCorrect entries and default file locations. 2. Resolves simulated software tasks using F1 Help search. 3. Adds frequently used commands to QAT. 4. Switches between Layout views.	• PC Workstations with Internet Access • MS Office Help & Support Documentation • Custom QAT Configuration Guide • Ref: Microsoft Learn Word Documentation • Ref: KNEC ICT Module 1 Practical Guide	• Oral Troubleshooting Quiz • Practical Toolbar Customization Check	Date: 16/09/2026 Reflections: Trainee effectively used Help features to solve assigned formatting challenges.
3	Process computeriz	Session 5: Creation & Manipulation of	By the end of the session,	1. Demonstrates table creation, grid	1. Inserts and populates a	• PC with MS Word /	• Practical Table	Date: 21/09/2026

	ed word document	Tables Subtopics: 1.3.1 Inserting tables (grid, dialog, draw table) 1.3.2 Working with tables (merge/split cells, cell alignment, formulas, table styles)	trainee should be able to: 1. Create tables using multiple insertion methods. 2. Modify row/column dimensions and merge cells. 3. Apply borders, shading, and table styles. 4. Insert basic formulas inside tables.	adjustments, and cell cell merging. 2. Explains table alignment, text orientation, and custom borders. 3. Illustrates calculating sums in Word tables. 4. Tasks trainee with creating a complex invoice table.	6x6 table with data. 2. Merges headers, adjusts padding, and applies custom shading. 3. Performs basic calculations (=SUM, =AVERAGE). 4. Formats data alignment within cells.	LibreOffice Writer • Sample Financial Data & Invoice Layouts • Step-by-Step Table Formatting Lab Sheet • Ref: PCAP / MOS Office Training Manuals • Ref: TVET CDACC Computer Operations Syllabus	Creation Task • Accuracy & Alignment Verification Checklist	Reflections: Table creation completed successfully. Formula integration in Word appreciated.
3	Process computerized word document	Session 6: Mail Merge Operations Subtopics: 1.4.1 Mail merge concepts & preparation 1.4.2 Main document & recipient data source setup 1.4.3 Inserting merge fields & previewing 1.4.4 Mail merge output (print/letters/envelopes)	By the end of the session, trainee should be able to: 1. Explain the mechanism of mail merge. 2. Create a data source table (Excel/Word). 3. Link main document with dataset. 4. Complete merge to generate personalized letters.	1. Explains mail merge logic (Main doc + Data source = Merged output). 2. Demonstrates Mail Merge Wizard & step-by-step custom merging. 3. Shows filtering/sorting recipient list. 4. Supervises practical mail merge task.	1. Creates a main letter document and recipient database (5+ contacts). 2. Inserts fields (FirstName, Address, Amount due). 3. Previews merged records for errors. 4. Executes merge to individual documents.	• Workstation Computers & Printer • Sample Recipient CSV/Excel Spreadsheets • Standard Letter Template Files • Ref: Microsoft Support Mail Merge Tutorials • Ref: KNEC Diploma ICT Module 1 Notes	• Individual Mail Merge Practical Exam • Output File Verification Check	Date: 23/09/2026 Reflections: Mail merge successfully executed without field matching errors.
4	Process computerized word document	Session 7: Word Processing Objects & Automatic Content Subtopics:	By the end of the session, trainee should be able to: 1. Insert and	1. Demonstrates object positioning, inline vs floating, text wrapping. 2. Shows caption	1. Inserts pictures and SmartArt diagrams into a document.	• Computers, Projector & Internet • High-Resolution	• Practical Document Production Task • Structural	Date: 28/09/2026 Reflections: Trainee

		1.5 Inserting graphical objects (pictures, shapes, SmartArt) 1.6.1 Inserting object captions & List of Figures 1.6.2 Heading styles & automatic Table of Contents	format images, shapes, and SmartArt. 2. Wrap text around graphics accurately. 3. Apply heading styles (Heading 1, 2, 3). 4. Generate automatic TOC and List of Figures.	insertion and figure numbering. 3. Explains how heading styles feed automatic Table of Contents. 4. Guides building an mini research paper layout.	2. Applies text wrapping (Square, Tight) and styling. 3. Tags headings with built-in styles. 4. Generates dynamic Table of Contents and List of Figures.	Practice Image Stock • Multi-Page Draft Project Document • Ref: Word Advanced Technical Writing Standards • Ref: TVET CDACC Level 5 Assessment Tool	Integrity & Layout Evaluation	generated TOC correctly. SmartArt diagram formatting handled well.
4	Process computerized word document	Session 8: Document Output, Printing & Continuous Assessment Subtopics: 1.7.1 Print setup, margins, paper size & orientation 1.7.2 Printer settings, duplex printing, print preview 1.7.3 Finalizing & exporting to PDF 1.8 CAT 1: Word Processing Module	By the end of the session, trainee should be able to: 1. Configure page orientation, paper size (A4/Letter), and margins. 2. Preview and troubleshoot print alignment. 3. Export documents to standard PDF format. 4. Demonstrate complete word processing mastery.	1. Demonstrates Print Preview features, collating, paper scaling. 2. Shows PDF export and document protection. 3. Administers Continuous Assessment Test (CAT 1 - Practical & Theory).	1. Adjusts document page setup parameters. 2. Conducts print preview checks to avoid orphan headings. 3. Exports final clean version to PDF. 4. Completes CAT 1 practical tasks under timed conditions.	• Computers connected to Network Printer • A4 Printing Paper Stock & Toner • CAT 1 Question Paper & Marking Scheme • Ref: Aberdare Tech Examination Guidelines • Ref: TVET CDACC Assessment Guidelines	• CAT 1 Practical Test Score (100% Scale) • Printed Document Quality Verification	Date: 30/09/2026 Reflections: CAT 1 completed. Trainee achieved high proficiency in layout formatting and printing.
5	Manipulate computerized spreadsheet	Session 9: Spreadsheet Concepts & Workbook Creation Subtopics: 2.1.1 Intro to	By the end of the session, trainee should be able to: 1. Define spreadsheet terminology.	1. Explains spreadsheet utility in business and data analysis. 2. Demonstrates interface component	1. Launches MS Excel / LibreOffice Calc. 2. Identifies and labels interface	• PC Workstations & High Resolution Projector • MS Excel 2019 / LibreOffice Calc	• Interface Identification Practical Quiz • Workbook Creation Verification	Date: 05/10/2026 Reflections: Interface layout mastered

		electronic spreadsheets 2.1.2 Elements of Excel/Calc window (ribbon, formula bar, name box, grid) 2.1.3 Worksheets vs workbooks 2.1.4 Creating & saving workbooks	2. Identify window elements. 3. Navigate worksheet grid (rows, columns, cells). 4. Create, save, and protect workbooks.	functions. 3. Guides navigation keyboard shortcuts (Ctrl+Home, Ctrl+Arrow keys). 4. Assigns workbook initialization exercise.	elements. 3. Navigates large grid datasets using shortcut keys. 4. Creates and saves a multi-sheet workbook structure.	• Keyboard Navigation Shortcut Sheet • Ref: Excel 2019 Bible by Michael Alexander • Ref: CDACC ICT Level 5 Module 2 Guide		quickly. Grid navigation performed fluently.
5	Manipulate computerized spreadsheet	Session 10: Cell Referencing & Worksheet Management Subtopics: 2.2.1 Cell referencing types: Relative (A1), Absolute (\$A\$1), Mixed (A\$1, \$A1) 2.2.2 Sheet editing: insert/delete/rename sheets, row height, column width, grid lines	By the end of the session, trainee should be able to: 1. Differentiate relative, absolute, and mixed references. 2. Apply \$ locking mechanisms in formulas. 3. Manage worksheet tabs (color, reorder, hide). 4. Adjust layout grid properties.	1. Explains mechanics of relative vs fixed cell references. 2. Demonstrates F4 shortcut key for absolute locking. 3. Illustrates tab operations (rename, recolor, copy sheet). 4. Provides tax/pricing calculation drill.	1. Constructs multiplication table using mixed references. 2. Applies absolute references to calculate sales tax across rows. 3. Renames, colors, and copies worksheets. 4. Adjusts row heights and column auto-fit.	• Workstation Computers with Excel • Reference Mechanics Practice Dataset • Lab Guide on Cell Addressing Techniques • Ref: Microsoft Excel Formulas & Functions Book • Ref: KNEC Computer Operations Notes	• Formula Audit Practical Exercise • Direct Observation of Reference Accuracy	Date: 07/10/2026 Reflections: Concept of absolute referencing (\$) fully grasped through practical drill.
6	Manipulate computerized spreadsheet	Session 11: Data Entry, Formatting & Types Subtopics: 2.2.3 Data entry rules & shortcuts 2.2.4 Data types: Text, Numeric, Date/Time, Currency, Percentage	By the end of the session, trainee should be able to: 1. Enter and edit data accurately. 2. Format data types correctly. 3. Utilize AutoFill handle for sequences.	1. Explains text alignment (left) vs numeric alignment (right) defaults. 2. Demonstrates Number formatting dialog (decimals, currency, custom dates). 3. Shows AutoFill capabilities (dates, months, custom	1. Enters complex dataset containing mixed data types. 2. Formats numbers as Kenyan Shillings (KES) currency and	• PCs & Projector • Raw Enterprise Data Entry Sheets • Custom Number Formatting Guide • Ref: Excel Data Analysis & Business	• Data Entry Speed & Accuracy Assessment • Format Application Audit	Date: 12/10/2026 Reflections: Data typing rules understood. Currency and percentage formatting correctly

		2.2.5 Data formatting, autofill & custom list creation	4. Build custom AutoFill series lists.	lists). 4. Assigns payroll data entry task.	percentages. 3. Generates date sequences using fill handle. 4. Creates custom auto-fill list for company departments.	Modeling (Wayne Winston) • Ref: TVET CDACC ICT Level 5 Guide		applied.
6	Manipulate computerized spreadsheet	Session 12: Standard Formulas & Built-in Functions Subtopics: 2.3.1 Syntax of formulas (=OPERATOR) 2.3.2 Mathematical & statistical functions (SUM, AVERAGE, COUNT, COUNTA, MIN, MAX) 2.3.3 Logical functions (IF, AND, OR) 2.3.4 Lookup functions (VLOOKUP, HLOOKUP)	By the end of the session, trainee should be able to: 1. Write clean formula syntax. 2. Apply core math & stat functions. 3. Formulate nested IF statements. 4. Execute VLOOKUP queries to combine data.	1. Explains formula syntax rules and error codes (#VALUE!, #N/A, #REF!). 2. Demonstrates SUM, AVERAGE, COUNTIF formulas. 3. Shows logical evaluation using IF and VLOOKUP table indexing. 4. Guides trainee through student grading analysis sheet.	1. Calculates total, mean, min, max for student marks. 2. Writes IF formulas to assign Pass/Fail and letter grades. 3. Uses VLOOKUP to retrieve student details from external master table. 4. Troubleshoots formula error messages.	• Computers with MS Excel • Sample Student Marks & Inventory Datasets • Formula Reference Quick Guide • Ref: MOS Excel 2019 Expert Study Guide • Ref: KNEC Diploma ICT Past Papers	• Practical Formula & Function Assessment • Logical Function Logic Check	Date: 14/10/2026 Reflections: VLOOKUP and IF statements successfully implemented . Excellent analytical progress.
7	Manipulate computerized spreadsheet	Session 13: Spreadsheet Formatting & Presentation Subtopics: 2.3.2.1 Typography & cell shading 2.3.2.2 Cell alignment & text	By the end of the session, trainee should be able to: 1. Apply visual cell styling and themes. 2. Wrap text and merge title blocks.	1. Demonstrates table aesthetic principles (contrast, subtle fills). 2. Shows Conditional Formatting setup (Greater Than, Highlight top 10%, Data Bars).	1. Applies cohesive color palette to financial data table. 2. Configures conditional formatting to highlight overdue	• Workstation PCs & Projector • Financial Report Raw Files • Design Guidelines for Corporate Spreadsheets • Ref:	• Visual Presentation Assessment • Conditional Formatting Verification	Date: 19/10/2026 Reflections: Conditional formatting highlights applied effectively. Clean

		wrapping 2.3.2.3 Custom borders & gridline control 2.3.2.4 Conditional formatting rules & data bars 2.3.2.5 Headers, footers & print titles	3. Build conditional formatting highlight rules. 4. Insert headers/footers with dynamic page numbers.	3. Demonstrates Page Layout header/footer insertion. 4. Tasks trainee to design executive financial dashboard sheet.	balances in red. 3. Sets repeating header rows for multi-page printing. 4. Inserts dynamic header (Date, File Path, Page X of Y).	Storytelling with Data by Cole Nussbaumer Knaflic • Ref: TVET CDACC Level 5 Practical Workbook		presentation achieved.
7	Manipulate computerized spreadsheet	Session 14: Data Visualization & Chart Generation Subtopics: 2.4.1.1 Chart types (Column, Bar, Line, Pie, Combination) 2.4.1.2 Creating charts from data series 2.4.1.3 Chart elements (titles, legend, data labels, axes) 2.4.1.4 Formatting and moving charts	By the end of the session, trainee should be able to: 1. Select appropriate chart type for data. 2. Insert 2D/3D charts into worksheets. 3. Add and format chart components. 4. Move charts to standalone chart sheets.	1. Explains data chart matching rules (Pie=Parts of whole, Line=Trends, Column=Comparison). 2. Demonstrates chart generation, element toggling, and trendlines. 3. Assigns sales trends chart project.	1. Selects data ranges and inserts Column and Line charts. 2. Customizes chart title, axis titles, data labels, and colors. 3. Creates a Pie chart showing departmental expenditure breakdown. 4. Moves chart to standalone sheet and formats layout.	• Workstation Computers • Multi-Year Business Sales Datasets • Chart Selection Guide Poster • Ref: Microsoft Excel 2019 Data Analysis and Business Modeling • Ref: CDACC ICT Level 5 Module Guidelines	• Chart Generation Practical Task • Data Visualization Accuracy Check	Date: 21/10/2026 Reflections: Proper chart selection demonstrated. Visual elements configured properly.
8	Manipulate computerized spreadsheet	Session 15: Sheet Printing, Page Setup & Exporting Subtopics: 2.4.1.4 Print area definition & fit-to-page scaling	By the end of the session, trainee should be able to: 1. Set explicit Print Areas. 2. Scale worksheets to fit	1. Demonstrates Page Break Preview and manual break dragging. 2. Explains Fit to 1 Page Wide scaling mode.	1. Defines specific print area on large workbook. 2. Adjusts scaling to fit report on single page	• PC Workstations connected to Laser Printer • Large Complex Datasets (50+ rows)	• Practical Sheet Printing Assessment • Print Output Quality Inspection	Date: 26/10/2026 Reflections: Fit-to-page setup executed flawlessly.

		2.4.1.5 Page setup (orientation, margins, print titles) 2.4.1.6 Gridlines & row/column header printing 2.4.1.7 Exporting sheets to PDF/CSV	on 1 page wide. 3. Toggle gridlines & row/column headings print visibility. 4. Export spreadsheets to CSV and PDF.	3. Shows how to enable gridlines in print output. 4. Demonstrates CSV export for database import.	width. 3. Enables gridlines and heading printing in Page Setup. 4. Exports dataset as PDF report and clean CSV file.	• Print Optimization Checklist • Ref: Excel Formatting & Printing Masterclass Guide • Ref: Aberdare Tech ICT Dept Rules		No fragmented table columns on print.
8	Manipulate computerized spreadsheet	Session 16: Comprehensive Spreadsheet Practical & Assessment Subtopics: 2.5 Integrated Spreadsheet Capstone Task 2.5.1 Multi-sheet model design 2.5.2 Complex formulas, lookup, charts & formatting 2.6 CAT 2: Spreadsheet Manipulation Module	By the end of the session, trainee should be able to: 1. Integrate data entry, formulas, and lookup functions. 2. Construct professional summary charts. 3. Apply formatting and print optimization. 4. Complete timed assessment independently.	1. Provides comprehensive business case scenario (Payroll & Sales Analysis). 2. Clarifies assessment requirements and time limits. 3. Monitors independent practical execution. 4. Evaluates completed workbooks against rubric.	1. Analyzes raw business data requirements. 2. Builds multi-sheet model with automated formulas (SUM, IF, VLOOKUP). 3. Generates summary chart dashboard. 4. Submits formatted printed report and digital .xlsx file for CAT 2.	• Workstation PCs & Exam Environment • CAT 2 Assessment Specification Booklet • Grading Rubric Sheets • Ref: TVET CDACC Level 5 Assessment Framework • Ref: KNEC Practical Assessment Standards	• CAT 2 Practical Examination Score (100% Scale) • Automated Formula Verification Check	Date: 28/10/2026 Reflections: CAT 2 completed with high accuracy. Excellent mastery of formulas and chart building.
9	Maintain computerized database	Session 17: Database Concepts & Requirements Gathering Subtopics: 3.1.1 Introduction to Relational Databases (DBMS vs RDBMS)	By the end of the session, trainee should be able to: 1. Define database terms (Table, Field, Record, Primary Key, Foreign Key).	1. Explains database fundamentals, data redundancy, and integrity rules. 2. Compares flat-file spreadsheets vs relational databases. 3. Presents	1. Identifies key entities and attributes in sample scenario. 2. Distinguishes primary keys and foreign key	• Workstation Computers & Projector • MS Access / LibreOffice Base / MySQL Workbench • Library System Case Study Document	• Oral Questioning on DBMS Concepts • User Requirements Analysis Sheet Check	Date: 02/11/2026 Reflections: Trainee clearly understood difference between flat files and

		3.1.2 Database organization (tables, records, fields, keys) 3.1.3 Collection of user requirements for database design	2. Explain benefits of RDBMS over spreadsheets. 3. Gather user requirements from business case.	business scenario (College Library System) to extract requirements. 4. Guides requirement list compilation.	relationships. 3. Documents functional user requirements for database system. 4. Participates in class discussion.	• Ref: Database System Concepts (Silberschatz, Korth, Sudarshan) • Ref: TVET CDACC ICT Level 5 DB Module		relational tables.
9	Maintain computerized database	Session 18: Database Schema & Entity Relationship Modeling Subtopics: 3.2 Designing computerized database schema 3.2.1 ERD models (Entities, Attributes, Relationships: 1:1, 1:M, M:N) 3.2.2 Relational models & Normalization basics (1NF, 2NF, 3NF)	By the end of the session, trainee should be able to: 1. Identify Entities, Attributes, and Relationships. 2. Construct Entity Relationship Diagrams (ERD). 3. Convert ERD diagrams into relational schemas. 4. Apply basic normalization principles.	1. Demonstrates ERD notation standard (Boxes, Ellipses, Diamonds/Lines). 2. Explains cardinality (One-to-Many, Many-to-Many). 3. Shows how to resolve Many-to-Many with junction tables. 4. Guides ERD drawing using Draw.io / Vision / Whiteboard.	1. Analyzes requirements to extract Entities (e.g., Student, Book, Borrowing). 2. Draws comprehensive ERD diagram with attributes and key constraints. 3. Converts ERD into relational table structures with FKs. 4. Normalizes tables to 3NF.	• PCs with Diagramming Software (Draw.io / Lucidchart) • ERD Modeling Stencils & Templates • Sample Database Design Scenarios • Ref: Fundamentals of Database Systems (Elmasri & Navathe) • Ref: KNEC Diploma ICT Database Systems Notes	• ERD Diagram Design Practical Review • Relational Schema Conversion Audit	Date: 04/11/2026 Reflections: ERD diagram constructed correctly. Junction tables introduced to resolve M:N relationships.
10	Maintain computerized database	Session 19: Database Objects Creation (Tables, Forms, Queries, Reports) Subtopics: 3.3.1 Table creation, data types & field properties	By the end of the session, trainee should be able to: 1. Create database tables with precise field data types. 2. Establish Referential	1. Demonstrates table creation in MS Access / Base design view. 2. Sets field properties (Field Size, Input Mask, Validation Rule). 3. Shows visual relationship window	1. Creates database tables (e.g., tblStudents, tblCourses, tblEnrollment). 2. Defines primary/foreign keys and	• Computers with MS Access / LibreOffice Base • Data Dictionary Specification Worksheet • Database Design	• Practical Database Creation Assessment • Referential Integrity & Object Setup Check	Date: 09/11/2026 Reflections: Table relationships and primary key constraints established

		3.3.2 Setting Primary Keys & Table Relationships 3.3.3 Creating Input Forms & Select Queries 3.3.4 Designing & formatting Reports	Integrity constraints. 3. Build user-friendly data entry Forms. 4. Design parameter Queries and summary Reports.	link setup. 4. Demonstrates Form Wizard, Query Design view, and Report Generator.	enforces Referential Integrity. 3. Builds dynamic entry Form with buttons. 4. Creates Select Query with criteria and formatted Report layout.	Template Files • Ref: Microsoft Access 2019 Bible by Michael Alexander • Ref: CDACC Level 5 DB Practical Guide		without errors.
10	Maintain computerized database	Session 20: Data Manipulation (CRUD) & Database Printing Subtopics: 3.4.1 Inserting records via Forms & Datasheet 3.4.2 Retrieving & filtering records (Queries) 3.4.3 Updating & deleting records safely 3.4.4 Printing tables, query results & formatted reports	By the end of the session, trainee should be able to: 1. Populate database tables accurately. 2. Perform SQL/GUI Update and Delete operations. 3. Filter records using multi-criteria queries. 4. Print formatted database reports.	1. Demonstrates record insertion, navigation, and validation error handling. 2. Explains action queries (Update, Delete, Append). 3. Shows page setup for printing wide database tables and reports. 4. Guides practical CRUD maintenance task.	1. Inserts 15 sample records into database tables using custom Form. 2. Executes criteria query to filter active records. 3. Performs record updates and tests referential integrity deletion blocks. 4. Configures print options and prints formatted database Report.	• PCs & Network Printer • Raw Enterprise Datasets for Entry • SQL & Query Criteria Quick Reference • Ref: Learning SQL by Alan Beaulieu • Ref: Aberdare Tech ICT Dept Exam Standards	• Practical Database CRUD Manipulation Task • Printed Database Report Audit	Date: 11/11/2026 Reflections: CRUD operations performed smoothly. Parameter queries worked as expected.
11	Prepare PowerPoint presentation	Session 21: PowerPoint Requirements, Slide Creation & Views Subtopics:	By the end of the session, trainee should be able to: 1. Analyze presentation	1. Explains visual communication principles and content structure. 2. Demonstrates theme selection,	1. Creates a 6-slide presentation structure based on given topic.	• PC Workstations & Digital Projector • MS PowerPoint 2019 /	• Slide Layout Practical Check • Presentation Structure Evaluation	Date: 16/11/2026 Reflections: Slide master edits applied

		4.1.1 Intro to presentation software & concepts 4.1.2 Target audience analysis & slide design principles 4.2.1 Slide layouts, themes & master slides 4.3 Presentation views (Normal, Slide Sorter, Reading, Slide Show)	requirements. 2. Apply slide design rules (6x6 rule, contrast). 3. Create presentation using master slide templates. 4. Navigate and utilize slide views effectively.	slide layout customization, and Master Slide editing. 3. Shows Slide Sorter view for reordering slides. 4. Assigns corporate presentation build.	2. Customizes Master Slide header/footer, logo position, and fonts. 3. Applies slide layout variations (Title, Two Content, Comparison). 4. Uses Slide Sorter to re-sequence content.	LibreOffice Impress • Slide Design Guidelines & Typography Poster • Ref: Presentation Zen by Garr Reynolds • Ref: TVET CDACC ICT Level 5 Presentation Guide		successfully across all slides. Good design choices.
11	Prepare PowerPoint presentation	Session 22: Animation, Transitions, Slide Objects & Printing Subtopics: 4.4 Slide transition effects & timing control 4.5.1 Custom text/object animation (Entrance, Emphasis, Exit) 4.5.2 Inserting tables, charts, multimedia (Audio/Video) 4.6 Presenter View setup & Handout printing	By the end of the session, trainee should be able to: 1. Apply seamless slide transitions. 2. Build custom object animations. 3. Embed tables, charts, and video clips. 4. Configure Presenter View and print audience handouts.	1. Demonstrates slide transition timing and sound effects. 2. Illustrates Animation Pane, trigger timing, and motion paths. 3. Shows embedding Excel charts and media files into slides. 4. Demonstrates printing Handouts (3/6 slides per page) and Presenter View setup.	1. Applies smooth fade transitions across slides. 2. Animates bullet items to appear sequentially on click. 3. Embeds a financial summary chart and video file into slides. 4. Configures Presenter view and exports 3-slide handout sheet to printer/PDF.	• PCs, Projector & Speaker System • Audio/Video Media Sample Stock • Excel Source Chart Files • Ref: Slide:ology by Nancy Duarte • Ref: KNEC Computer Operations Study Manual	• Animated Presentation Assessment • Handout Output Verification	Date: 18/11/2026 Reflections: Animations set appropriately without overwhelming layout. Handout printing executed.
12	Manipulate graphic application	Session 23: Graphic Design Requirements & Publication Setup	By the end of the session, trainee should be able to: 1. Differentiate	1. Explains graphic design fundamentals, color modes (RGB vs CMYK), and	1. Launches graphic application (MS Publisher / Inkscape).	• PCs with MS Publisher / Inkscape / CorelDraw • High-	• Graphic Setup Practical Exercise • Canvas	Date: 23/11/2026 Reflections: Page layout

		Subtopics: 5.1.1 Intro to desktop publishing & graphic terms (vector vs raster, DPI, CMYK/RGB) 5.1.4 Publication templates (flyers, brochures, banners, business cards, certificates) 5.2.1 Page setup, margin rules, guides, rulers & grids	raster (PNG/JPEG) vs vector formats. 2. Select appropriate publication templates. 3. Configure page setup, bleed, orientation, and margin guides. 4. Plan multi-page publications.	resolution. 2. Demonstrates MS Publisher / Canva / CorelDraw / Inkscape setup. 3. Shows guide positioning, column grid setup, and template customization. 4. Tasks trainee to set up a tri-fold brochure canvas.	2. Sets paper dimensions, margin guides, and bleed lines for a brochure. 3. Selects color palette adhering to CMYK standard. 4. Places design placeholders on layout canvas.	Resolution Image Library & Graphics Assets - Graphic Design Fundamentals Handbook - Ref: The Non-Designer's Design Book by Robin Williams - Ref: TVET CDACC Graphic Design Syllabus	Layout & Guide Precision Check	and guide lines set correctly for tri-fold publication template.
12	Manipulate graphic application & Collaboration	Session 24: Graphic Publishing, Document Production & Online Collaboration (Final Assessment) Subtopics: 5.2.2 - 5.3.3 Graphic publishing (textboxes, shapes, layer ordering, high-res export & printing) 6.1 - 6.3 Document production (scanning, optical character recognition - OCR, duplication) 7.1 - 7.4 Online & Mobile Collaboration (Google Drive, MS Teams, OneDrive, real-time co-authoring & mobile sync) 8.0 Final	By the end of the session, trainee should be able to: 1. Publish formatted marketing publication. 2. Operate scanner/duplicator hardware & OCR software. 3. Utilize online cloud collaboration tools (Drive/Teams). 4. Synchronize mobile and desktop files. 5. Complete comprehensive practical summative exam.	1. Demonstrates visual element layering, wrapping, and PDF export for high-res printing. 2. Demonstrates document scanner setup, resolution tweaking, and OCR scanning. 3. Demonstrates real-time document sharing, permissions, and co-authoring on Google Docs / MS Teams. 4. Administers Final Integrated Practical Examination (CAT 3) covering all 7 unit outcomes.	1. Finalizes design of event flyer/brochure and exports ready-to-print file. 2. Scans paper document into editable text using scanner software. 3. Shares project folder via Google Drive/OneDrive, setting edit permissions. 4. Executes real-time collaborative editing exercise on cloud file. 5. Completes	- Workstation Computers & Network Color Printer - Flatbed Document Scanner & OCR Software - Internet Connectivity & Cloud Accounts (Google/Microsoft) - Comprehensive CAT 3 Exam Booklet & Marking Rubric - Ref: CDACC Unit Assessment Tool - Unit 0611 351 02A - Ref: KNEC National Examinations	- CAT 3 Comprehensive Practical Exam Score - Online Collaboration Task Verification - Printed Publication Quality Review	Date: 25/11/2026 Reflections: Final integrated assessment successfully completed. All learning outcomes verified.

		Comprehensive Integrated Assessment (CAT 3)			comprehensive CAT 3 practical exam covering Word, Excel, Access, PowerPoint, Desktop Publishing, and Collaboration.	Syllabus		
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